Pre-analysis plan for #manydaugthers study by Agne Kajackaite and Anastasia Danilov
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Hypothesis	Analysis plan
H1. *Having daughters leads to stronger support for gender equality (as measured by attitudes).* That is, denoting X as a measure of support for gender equality, (X for respondents with daughters) > (X for respondents without daughters).	Only use data from year 2018, or the most recent available year for the dependent variable. 1) **Attitudes toward working mothers:** Children below the age of 6 suffer if their mother works. (1 compl. disag. -7 fully agr., reversed) 2) **Attitudes toward household equality:** It's best if the man and the woman work the same amount so they can share the responsibility for taking care of the family and household equally. (1 compl. disag. -7 fully agr.) 3) **Average score of attitudes toward women's financial independence:** Even a married woman should have a paid job so that she can be financially independent. (1 compl. disag. -7 fully agr.) Having a job is the best way for a woman to be independent (1 compl. disag. -7 fully agr.) If a woman earns more money than her partner, this inevitably leads to problems (1 compl. disag. -7 fully agr., reversed). 4) **Average score of male/female power distribution:** At home, the husband should have the final say. (1 compl. disag. -7 fully agr., reversed) It's best if the man and the woman work the same amount so they can share the responsibility for taking care of the family and household equally. (1 compl. disag. -7 fully agr.) 5) **Attitudes towards equality in education:** For parents, vocational training or higher education for their sons should be more important than vocational training or higher education for their daughters. (1 compl. disag. -7 fully agr., reversed) Empirical strategy:

	Ordinary Least Squares (OLS) regression with robust standard errors (see the main specification below). Interpretation: The positive and statistically significant coefficient for *Daughter* (at the 5%-level) suggests that having a daughter is associated with more progressive attitudes toward gender equality.
H2. *Having daughters leads to more left-leaning political preferences.* That is, denoting Y as a measure of left-leaning political preferences, (Y for respondents with daughters) > (Y for respondents without daughters).	2019 as the most recent available year: **Political attitude:** In politics people often talk about "left" and "right" when it comes to characterize different political attitudes. If you think about your own political views: Where would you place yours? (0 far left, 10 far right) Empirical strategy: OLS regression with robust standard errors (see the main specification). Interpretation: - The negative and statistically significant coefficient for *Daughter* (at the 5%-level) indicates that having a daughter is associated with more left-leaning political orientation.
H3. *Having daughters leads to more gender equality in the household (**as measured by behaviors or outcomes**).* That is, denoting Z as a measure of gender equality in the household, (Z for respondents with daughters) > (Z for respondents without daughters).	**Financial decisions** (year 2018): - Who has the last word in your relationship / marriage when making important financial decisions? (1 only man; 0 otherwise) Empirical strategy: Linear Probability Model, i.e. OLS with robust standard errors (see the main specification). Interpretation: - The negative and statistically significant coefficient for *Daughter* (at the 5%-level) indicates reduced male dominance in financial household decisions, suggesting greater gender equality within households.

	Time invested in childcare and household responsibilities (year 2020): - Natural logarithm of total number of daily hours spent on housework, childcare, and household errands. Empirical strategy: OLS regression with robust standard errors (see the main specification). Separate regressions are estimated for male and female respondents. Interpretation: - In the male sample, a positive and statistically significant coefficient for *Daughter* (at the 5% level) suggests that having a daughter leads men to invest more time in childcare and household tasks, indicating increased gender equality. - In the female sample, a negative and statistically significant coefficient for *Daughter* (at the 5% level) suggests that having a daughter is associated with women spending less time on these tasks, also reflecting greater gender equality in the household.
H4. *Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).* That is, denoting X as a measure of support for gender equality,	All dependent variables are as in H1, but the model includes an interaction term *Daughter x Male respondent,* where Male respondent is equal one if the respondent is a man and zero if the respondent is a woman. Interpretation: The positive and statistically significant coefficient of *Daughter x Male respondent* (at the 5%-level) indicates that having a daughter leads *to* a greater increase in support of gender equality among men than among women.

Empirical Strategy:

$$Y_i = \alpha + \beta_1 * Daughter_i + \mathrm{X}_i'\delta + \epsilon_i$$

Y_i is the outcome variable of interest for individual i. The outcome variables for H1, H2, and H3 are z-scores normalized at the mean value and divided by standard deviation.

Main independent variable:

- gender of the first-born child: $Daughter_i$ is equal to 1 if the first-born child is a daughter.

X_i is a vector of control variables:

- male respondent (equals one if the respondent is a man and zero if the respondent is a woman),
- age of the respondent,
- the number of (other) daughters, i.e. daughters among the second and later born children,
- the number of children,
- race of the respondent (dummy equals one if White-Caucasian vs. zero if not),
- age of the first-born child at the point of the survey,
- highest educational level achieved by the respondent (linear value, 0 = without school degree, 1 = school degree, 2 = professional degree (*Ausbildung*), 3 = university degree)
- marital status (dummy, equals one if married or living with a partner, zero otherwise),
- income,
- political party affiliation,
- religion,
- employment status (linear value, 0 not employed, 1 part-time, 2 full-time, 3 self-employed),
- east/west Germany (dummy).

If some of the variables are not available, they will be excluded from the analysis.

The error term is represented by ϵ_i. We use robust standard errors.

Sample exclusions:

- always take the latest year available for the dependent variable,
- individuals with 1, 2, 3, 4, or 5 children (i.e. only parents with less than 6 kids),
- exclude the families with the first -born twins or triplets,
- exclude “probably bi/homosexual” as in *ppath/sexor: Sexual Orientation*,
- include only respondents who identify either as a man or a woman,
- include only respondents of age 25-60,
- include only respondents who were born in Germany.